

Volume is up 70%. Profit per transaction is down 33%.

We grew by adding cheap transactions, not profitable ones. 88.6% of the decline is a shift in what we process, not a change in how we price or how well we run.

\$1.61 → \$1.08

Contribution per successful transaction, Jan → Dec

88.6%

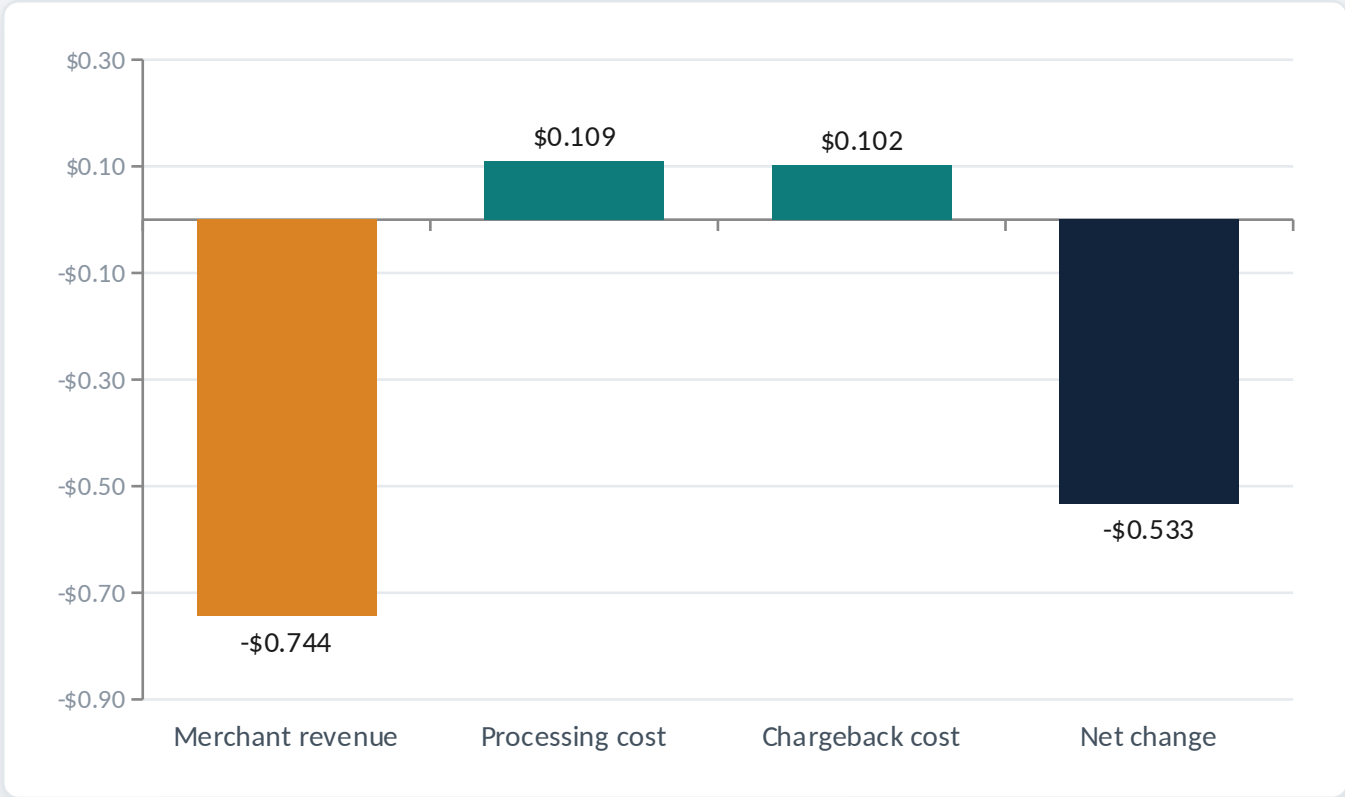
of the fall is ticket-size mix, not rate deterioration

39.3%

of the fall comes from over-\$500 payments losing share

The evidence: revenue per transaction collapsed, cost did not

Bridge from January to December contribution per successful transaction (USD)



1 Mix, not rate

Ticket-size mix explains $-\$0.472$ of the $-\$0.533$ move. Decomposition residual is exactly zero.

2 Small tickets took over

Under-\$25 share rose 15.0% $\rightarrow$ 33.7%. Over-\$500 share fell 10.2% $\rightarrow$ 7.0%.

3 Three hypotheses disproved

Pricing (no plan fell), fraud (-0.014pp), chargebacks ($\$0.25 \rightarrow \0.15) are all ruled out.

Data trust: 26 checks run, 10 failed. Two joins would have inflated volume 11.9 $\times$ and 1.8 $\times$ if left un-aggregated. 3,038 rows quarantined and fully reconciled.

Recommendation: price the floor, route the floor, defend the top

Small tickets are not the problem. Small tickets priced and routed like large ones are.

PRICE

Fee floor of \$0.35 on FLAT_2_0

DIGITAL_GOODS / AE went from +\$0.227 to – \$0.167 per transaction. A 2% fee on a \$14 ticket cannot cover a \$0.53 processing cost.

Returns the small-ticket cohort to positive contribution

ROUTE

Send sub-\$90 volume to PROV_NORTH

Under \$25, Orbit costs \$0.53 against North's \$0.14. Orbit is the right rail above \$500, the wrong one below it.

About \$0.30 per transaction on ~34% of volume

DEFEND

Protect over-\$500 volume and renegotiate Orbit

Large tickets improved their own economics while losing share. Orbit's fixed fee rose 50.6% mid-year with no repricing on our side.

Halts the 39.3% of the decline that mix caused

What we are not recommending: a fraud-model rollback or a chargeback programme. Both were tested and neither moved the KPI.

The next 60 days

Five actions, each with a named owner and a metric that tells us within the window whether it worked

#	Action	Owner	Success metric	Window
1	Route sub-\$90 transactions to PROV_NORTH	Head of Payments	Cost per successful txn by ticket band	Days 1–20
2	Apply a \$0.35 per-transaction fee floor on FLAT_2_0	CFO	Contribution per txn, DIGITAL_GOODS SG and AE	Days 1–30
3	Renegotiate the PROV_ORBIT fixed fee	Procurement	Fixed fee recorded in route_cost	Days 10–60
4	Protect large-ticket volume in TRAVEL and ELECTRONICS	Commercial	Share of successful volume above \$500	Days 15–60
5	Fix the wallet integration dropping merchant_id	Data Head	Missing merchant_id rate by channel	Days 20–45

Day-60 test: contribution per successful transaction stops falling and recovers toward \$1.30, with the under-\$25 cohort back above zero. If it has not moved, actions 1 and 2 were not the binding constraint and we re-open the pricing-plan analysis.

What could make this wrong

The analysis has four soft spots. Each has a check that would surface it before it costs us.

Merchant churn

A fee floor on SME digital-goods accounts may push some to a competitor. Contribution gains could be offset by lost volume.

Track churn count weekly on FLAT_2_0 accounts; cap the rollout at the top 50 merchants first.

Capacity at North

The routing shift assumes PROV_NORTH can absorb ~34% more volume at the quoted rate.

Confirm a volume commitment and re-quote before moving traffic.

Blind spot in the data

2.1% of wallet transactions carry no merchant_id, so cohort figures exclude them. 10 of 26 quality checks failed.

Action 5 restores attribution; the decomposition is re-run once it lands.

Latency is real but small

The MERIDIAN incident cut success from 90.1% to 83.3% (p95 852 → 5,663ms) and lost ~698 transactions. It hurts volume, not unit economics.

Fixing it does not move this KPI. It belongs on the reliability roadmap, not this plan.

Bottom line: the profit fall is a mix problem we created by growing into the cheapest tickets without pricing or routing for them. Both are fixable inside 60 days.

How the number is built

One transaction, one row. Every figure in this deck traces back to the same grain.

1

11 source files

Payments, settlements, refunds, chargebacks, FX rates, route costs, risk snapshots, merchants, customers, pricing plans.

2

Pre-aggregate, then join

Every one-to-many source is collapsed to transaction grain before it touches the fact table. This is what prevents the 11.9× inflation.

3

fact_transaction

One row per payment attempt. contribution = merchant revenue – processing cost – chargeback cost, all in USD at the settlement rate.

4

Exact decomposition

$\Delta c = \Sigma(w_1 - w_0) \cdot c_0 + \Sigma w_1 \cdot (c_1 - c_0)$. Mix term plus within term, residual 0.

Decisions that shape every number

- Date basis is the transaction date, not settlement date — a settlement-dated KPI would smear the incident across two months.
- FX uses the SETTLEMENT rate type, not MID. MID overstates revenue.
- Merchant risk band is joined as-of the transaction date, not the latest snapshot.
- Money is held as integer minor units until the final division.
- Rows with a missing merchant keep a portfolio-average revenue rate so the decomposition stays exact.

Data trust: what we found before we trusted the number

26 checks across completeness, validity, consistency, uniqueness, timeliness and accuracy.

10 / 26

checks failed

11.92×

volume inflation if the risk snapshot were joined naively

1.79×

inflation from the fraud-signal join

0

residual after quarantine reconciliation

Defect found	Dimension	Rows	What we did about it
Duplicate settlement rows for one payment	Uniqueness	2,668	Deduplicated on payment_id before the join; kept the earliest settlement
Negative gross amounts on captured payments	Validity	370	Quarantined — flagged, excluded from the KPI, still counted in the reconciliation
Missing merchant_id on wallet transactions	Completeness	2.1%	Kept as an "(unknown)" segment so the decomposition stays exact; fix is action 5
Fraud score v3.0 emits 0–100, v2 emits 0–1	Consistency	all v3 rows	Normalised to a common scale — this alone disproved the fraud hypothesis
Refunds and chargebacks are one-to-many	Consistency	—	Pre-aggregated to transaction grain before joining

Reconciliation: 3,038 excluded rows = 2,668 duplicate settlements + 370 impossible negatives + 0 unexplained. Nothing disappears silently.

What each stakeholder believed, and what the data says

Every hypothesis tested with mix held constant, so a segment shift cannot masquerade as a rate change.

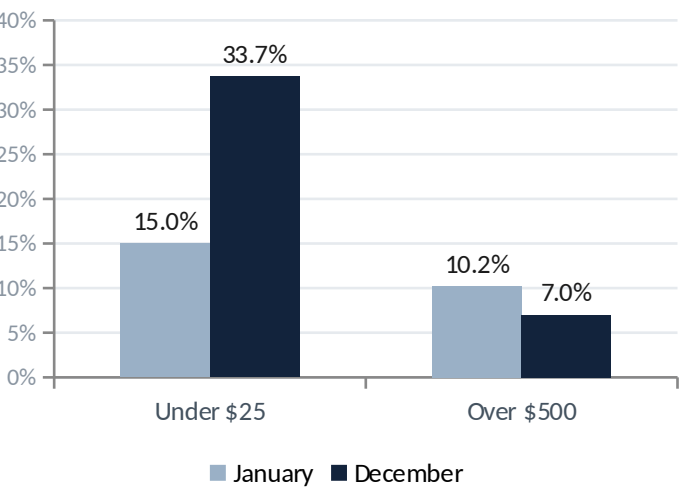
Claim	Who	Test	Result	Verdict
Merchants negotiated us down on price	CFO	Contribution rate by pricing plan, mix held constant	No plan fell. FLAT_2_0 rose 0.2247pp.	DISPROVED
The new fraud model is rejecting good payments	Risk Head	Decline rate on a normalised score scale	-0.014pp — declines fell slightly	DISPROVED
Chargebacks are eating the margin	Risk Head	Chargeback cost per successful txn	\$0.2515 → \$0.1495	DISPROVED
Provider latency is destroying conversion	Head of Payments	p95 latency and success rate, MERIDIAN	852 → 5,663ms; 90.1% → 83.3%; ~698 txns lost	REAL, WRONG KPI
FX movement is dragging revenue	—	FX impact per txn at the settlement rate	-\$0.01 → -\$0.09 (about 16% of the fall)	SECONDARY
Growth changed what we process	Our finding	Mix / within decomposition on ticket band	-\$0.472 mix vs -\$0.061 within	CONFIRMED

Three of the four hypotheses in the room were wrong. Acting on any of them would have spent budget without moving the KPI — which is the case for doing the decomposition before the workshop, not after it.

The mechanism, in three numbers

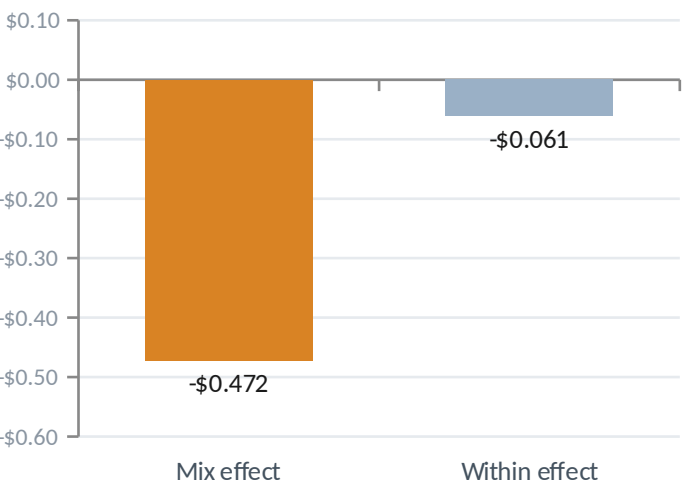
Why a percentage fee on a \$14 ticket cannot carry a fixed processing cost.

1. The mix moved



Share of successful transactions by ticket band.

2. Mix, not rate



88.6% of the fall is the weights moving, not segment economics worsening.

3. The cost floor bites

\$0.53
PROV_ORBIT cost on an under-\$25 payment

\$0.14
PROV_NORTH cost on the same payment

+50.6%
rise in Orbit's fixed fee, mid-year

+\$0.227 → -\$0.167
DIGITAL_GOODS / AE contribution per txn

Orbit is not a bad provider — it is the cheapest rail above \$500 at \$3.23 per transaction. We routed small tickets down a pipe priced for large ones, then let its fixed fee rise without repricing.

What this analysis does not tell you

Stated plainly, so nobody discovers them in month two.

Elasticity is unmeasured

We know a fee floor lifts contribution per transaction. We do not know how many merchants leave when it lands — there is no historical price change in the data to estimate from.

One year, no seasonality baseline

January and December are the comparison points. With a single year we cannot separate a seasonal ticket-mix pattern from a structural one.

Provider cost is a contracted rate, not an invoice

Costs come from route_cost, which is the rate card. Any volume rebate or true-up settled outside it is invisible here.

Attribution gap on wallet volume

2.1% of wallet transactions have no merchant, so merchant-level and cohort-level figures are computed on the remainder.

Causality is inferred, not experimental

The decomposition is exact arithmetic, not a controlled test. Actions 1 and 2 should launch on a merchant subset so the effect is measurable.

Next data we would ask for: merchant churn history, a volume-committed quote from PROV_NORTH, and provider invoices to reconcile against the rate card.